拥挤的苍穹

The Crowded Heavens



两颗卫星相撞，为什么多年后仍要避让？

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英文原文

On 10 February 2009, nearly 790 kilometres above Siberia, two satellites slammed together at about 42,000 kilometres an hour. Iridium 33 belonged to an American satellite-phone constellation; Kosmos 2251, a Russian military satellite, had been derelict since about 1995. Both shattered in the first accidental collision between two intact satellites.

Neither craft swerved. The Russian hulk, lacking propulsion, could not. A public collision-warning service had forecast a narrow miss of about 584 metres, yet the predicted encounter was not among the week's closest; such warnings were frequent and imprecise. One owner's discarded relic demolished another's working asset.

The wreckage, however, endangered everyone else. The impact flung out more than 2,000 catalogued fragments and a multitude of others too small to pinpoint. At that height air drag is feeble, so many would linger for decades. In January 2012, the International Space Station fired its thrusters to dodge an Iridium 33 remnant.

In June 1978, two NASA scientists envisaged a vicious spiral, later named the Kessler syndrome: each collision would scatter fragments, raising the odds of subsequent crashes. The 2009 crash illustrated that premise without triggering a runaway chain reaction. By the end of 2024, surveillance networks tracked about 40,000 objects, only some 11,000 of them functional.

Such a cascade would unfold over decades, though, and is not inevitable; the risk can be mitigated. Operators now routinely manoeuvre satellites out of harm's way. In September 2022, America's Federal Communications Commission, which licenses satellites, mandated disposal of low-orbit craft within five years of retirement instead of 25. Years after the crash, spacecraft whose owners had no part in it kept swerving around its splinters.

中文译文

2009年2月10日，西伯利亚上空近790公里处，两颗卫星以约每小时4.2万公里的相对速度猛撞在一起。铱星33号属于美国的一个卫星电话星座；俄罗斯军用卫星宇宙2251号则约从1995年起便已失效。两星双双解体，这是两颗完整卫星之间首次发生意外相撞。

两颗卫星都没有转向避让。俄方那颗失效卫星缺乏推进能力，无法躲开。一个公开的碰撞预警服务曾预测，两星会相距约584米擦过，但这次预计交会还不在当周距离最近的几次之列；这样的预警既频繁，又不精确。一方丢弃的旧物，摧毁了另一方仍在工作的资产。

碎片却危及所有其他航天器。撞击抛出了两千多块已编目碎片，还有大量小得无法精确定位的碎片。那个高度空气阻力很弱，许多碎片会在轨道上滞留数十年。2012年1月，国际空间站启动推进器，避开了一块铱星33号的残片。

1978年6月，两位美国航天局科学家设想了一种后来被称为“凯斯勒综合征”的恶性循环：每次碰撞都会散出碎片，提高后续相撞的概率。2009年的事故展示了这一机制的基础，却没有引发失控的连锁反应。截至2024年底，监测网络跟踪着约4万个在轨物体，其中只有约1.1万个正常运行。

不过，这样的连锁过程会历时数十年，并非不可避免，风险也能减轻。卫星运营方如今会经常调整卫星轨道，避开危险。2022年9月，负责颁发卫星许可的美国联邦通信委员会要求，其监管的低轨航天器在退役后五年内完成处置，而不再是25年。事故多年后，那些所有者与此事毫无关系的航天器，仍要一再转向躲避当年的碎片。

核心词汇

先记这22项。连着下方的原文短语一起回想。

derelict adj.
已废弃失效的
had been derelict since about 1995

collision n.
碰撞
the first accidental collision

propulsion n.
推进力；推进能力
lacking propulsion

narrow miss 名词短语
险些相撞；近距离擦过
a narrow miss of about 584 metres

imprecise adj.
不精确的
frequent and imprecise

discard v.
弃置
One owner's discarded relic

asset n.
资产
another's working asset

wreckage n.
残骸；碎片
The wreckage, however

catalogue v.
编目；登记
more than 2,000 catalogued fragments

pinpoint v.
精确定位
too small to pinpoint

air drag 名词短语
空气阻力
air drag is feeble

linger v.
滞留；迟迟不消失
many would linger for decades

dodge v.
躲避
to dodge an Iridium 33 remnant

envisage v.
设想
scientists envisaged a vicious spiral

subsequent adj.
后续的
the odds of subsequent crashes

premise n.
前提；基本设想
illustrated that premise

trigger v.
引发
without triggering

chain reaction 名词短语
连锁反应
a runaway chain reaction

surveillance n.
监视；监测
surveillance networks tracked

functional adj.
正常运行的
11,000 of them functional

inevitable adj.
不可避免的
is not inevitable

disposal n.
处置
mandated disposal of low-orbit craft

文中词汇与搭配

补充词汇共36项。保留每个表达在本文中的意思。

slam v.
猛撞

belong to 动词短语
属于

constellation n.
卫星星座；卫星群

shatter v.
粉碎；解体

accidental adj.
意外的

swerve v.
突然转向；转向避让

hulk n.
废弃的大型躯壳

encounter n.
交会；相遇

frequent adj.
频繁的

relic n.
遗留旧物

demolish v.
摧毁

endanger v.
危及

impact n.
撞击

fling out 动词短语
抛出；甩出

a multitude of 名词短语
大量的

feeble adj.
微弱的

space station 名词短语
空间站

fire v.
启动；点火

文中词汇与搭配 续

补充词汇共36项。保留每个表达在本文中的意思。

thruster n.
推进器

remnant n.
残片；遗留部分

vicious spiral 名词短语
恶性循环

syndrome n.
综合征

scatter v.
散布；使散开

raise the odds of 动词短语
提高……发生的概率

illustrate v.
例证；说明

runaway adj.
失控的

unfold v.
逐步展开

mitigate v.
减轻；缓解

routinely adv.
例行地；经常地

manoeuvre v.
操纵；调整运行轨迹

out of harm's way 介词短语
避开危险；在安全处

mandate v.
强制要求

retirement n.
退役

instead of 介词短语
而不是；代替

have no part in 动词短语
与……无关；未参与

splinter n.
碎片

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 the first accidental _____ between two intact satellites.

02 The Russian hulk, lacking _____, could not.

03 a multitude of others too small to _____.

04 and is not _____; the risk can be mitigated.

文章主线

- 相撞 2009年，两颗完整卫星首次意外相撞：失效的宇宙2251号撞上仍工作的铱星33号。
- 预警 公开预警频繁且不准，预计两星会隔约584米擦过，并非当周最近交会；失效卫星也无法避让。
- 现状 碰撞留下两千多块已编目碎片；截至2024年底，全球跟踪约4万个在轨物体，只有约1.1万个正常运行。
- 风险 高空阻力小，碎片可滞留数十年，又增加碰撞风险；此次未触发失控连锁反应，恶性循环并非必然。
- 应对 卫星运营方加强避让；美国联邦通信委员会把其监管低轨卫星的退役后处置期限，从25年缩至5年。

参考答案　1 collision 2 propulsion 3 pinpoint 4 inevitable